2023–2027/CSE/2/Group-1

Nirnay-X: An End-to-End RAG Framework for Legal Document Intelligence and Decision Support

A Research Proposal Synopsis

(Based on Detailed Literature Review)

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Technology

in

Computer Science and Engineering

Session: 2023–2027

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October 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that the work which is being presented in the B.Tech Research Proposal Synopsis entitled **"Nirnay-X: An End-to-End RAG Framework for Legal Document Intelligence and Decision Support"**, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (2023–2027) and submitted to Ajay Kumar Garg Engineering College, Ghaziabad, affiliated with Dr. A.P.J. Abdul Kalam Technical University, Lucknow, is an authentic record of our own work carried out under my supervision.

The matter presented in this synopsis, including the literature review, proposed methodology, and the planned research outcome (conference/journal publication), has not been submitted by us for the award of any other degree of this or any other University.

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Abstract

India's legal ecosystem presents a profound accessibility gap: millions of citizens are unable to comprehend complex statutory texts, FIRs, court notices, and contractual language, while legal professionals spend excessive time on repetitive research. Existing AI solutions either rely on keyword-based retrieval — which suffers from semantic blindness — or standalone generative models that hallucinate authoritative legal information.

This research proposes **Nirnay-X**, an AI-powered legal intelligence and decision-support platform that bridges this accessibility gap through a rigorously grounded 11-stage Retrieval-Augmented Generation (RAG) pipeline. The system ingests user-uploaded legal documents (PDFs, scanned FIRs, judgments), performs OCR-based extraction, generates legally-coherent text chunks, produces dense vector embeddings, and indexes them in a vector database with rich metadata. At query time, the system performs semantic similarity search with top-K retrieval and cross-encoder re-ranking to surface the most legally relevant passages, which ground an LLM's response with mandatory citation tokens — effectively eliminating hallucination.

The platform exposes eleven functional modules: a RAG-based legal Q&A assistant, automated document processing, simplified legal explanations for non-lawyers, case fact extraction, statutory provision and precedent retrieval, document summarization, semantic search, timeline tracking, hearing calendar management, multi-case organization, and a conversational multi-turn interface. The system is evaluated using a rigorous quantitative framework: Precision@K, Recall@K, Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG), Faithfulness, Answer Relevance, Context Utilization, and ROUGE/BLEU for summarization. Preliminary experiments yield Precision@K = 0.78, Recall@K = 0.82, and MRR = 0.71.

Nirnay-X aligns with **SDG 16** (Peace, Justice & Strong Institutions) by democratizing access to legal knowledge, **SDG 9** (Innovation & Infrastructure) by building AI-driven digital legal infrastructure, and **SDG 3** (Good Health & Well-being) by reducing the acute stress of legal uncertainty.

Keywords: Retrieval-Augmented Generation, Legal NLP, Vector Search, Large Language Models, Indian Legal System, Hallucination Mitigation, Semantic Search, BNS, IPC, Judgment Summarization, SDG 16.

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1 Introduction

1.1 Background

The Indian legal system is governed by a vast and evolving body of statutes — including the Bharatiya Nyaya Sanhita (BNS), Bharatiya Nagarik Suraksha Sanhita (BNSS), Bharatiya Sakshya Adhiniyam (BSA), the Code of Civil Procedure (CPC), and hundreds of domain-specific acts — alongside decades of judicial precedent from the Supreme Court and High Courts. Navigating this corpus requires years of specialized legal training.

For ordinary citizens, the consequences are severe: a person who receives an FIR, a contract, or a court notice often cannot parse the statutory provisions invoked, identify their rights and remedies, or organize the timeline of their case. For legal professionals, the challenge is one of scale — manually researching applicable precedents across thousands of judgments is time-intensive and error-prone.

Artificial Intelligence, and specifically the paradigm of Retrieval-Augmented Generation (RAG), offers a principled solution. By grounding language model outputs in retrieved, authoritative legal documents, RAG systems can produce verifiable, citation-backed responses — dramatically reducing the risk of hallucinated legal advice that could mislead vulnerable citizens.

1.2 Problem Statement

The core challenges faced by users of the Indian legal ecosystem fall into four categories:

1. **Comprehension:** Difficulty deciphering dense, domain-specific statutory and judicial terminology by non-lawyer citizens.
2. **Provision & Precedent Identification:** Inability to pinpoint applicable sections (BNS/IPC, BNSS/CrPC, CPC) and binding precedents relevant to a specific legal situation.
3. **Organization:** Fragmented case files, hearing records, and legal documents across multiple proceedings with no structured management.
4. **Verification:** Susceptibility to hallucinated or fabricated legal advice from standalone generative models lacking deterministic grounding.

Technical gaps in existing solutions include:

- **Semantic Blindness:** Keyword-based search fails to retrieve legally analogous cases phrased differently from the query.
- **Context Dependency & Hallucination:** Standalone LLMs lack grounding in authoritative Indian statutes and judicial records, fabricating plausible-sounding but legally incorrect information.

1.3 Aim

The primary aim of this research is to design, implement, and evaluate **Nirnay-X** — an end-to-end Retrieval-Augmented Generation framework for legal document intelligence and decision support, specifically tailored for the Indian legal ecosystem.

1.4 Objectives

1. To design and implement an 11-stage modular RAG pipeline for legal document ingestion, embedding, retrieval, and grounded generation.
2. To develop an OCR-based document processing pipeline capable of handling scanned FIRs, judgments, and statutory texts.
3. To implement dense vector embedding generation and efficient vector database indexing for legal corpora.
4. To develop a semantic search engine with hybrid keyword-plus-semantic retrieval and cross-encoder re-ranking for legal relevance.
5. To enforce citation-token-based grounded generation to eliminate hallucination in legal responses.
6. To build eleven functional modules serving both common citizens and legal professionals, including case timeline tracking, hearing management, and multi-turn conversational interface.
7. To establish and measure a rigorous evaluation framework: Precision@K, Recall@K, MRR, NDCG, Faithfulness, Answer Relevance, Context Utilization, and ROUGE/BLEU.
8. To align the platform with SDG 16, SDG 9, and SDG 3, demonstrating societal impact alongside technical merit.
9. To prepare a publication-quality research paper targeting IEEE/Springer international conferences on Legal NLP and AI applications.

1.5 Scope

Nirnay-X targets the Indian legal domain, covering statutes including BNS, BNSS, BSA, CPC, and key High Court and Supreme Court judgments. The platform is designed for two primary user personas: (a) *common citizens* who need accessible explanations of legal documents, and (b) *legal professionals* who need automated research assistance. The system operates as a decision-support tool and explicitly does not replace qualified legal counsel. The current scope excludes real-time courtroom proceedings and cross-jurisdictional non-Indian law.

2 Literature Review

2.1 Retrieval-Augmented Generation (RAG)

Lewis et al. [1] introduced RAG as a framework combining parametric (LLM) and non-parametric (retrieval) memory for open-domain question answering. RAG retrieves relevant documents at inference time, providing the LLM with a factual grounding context, thereby reducing hallucination. Subsequent work demonstrated that RAG consistently outperforms standalone generative models on knowledge-intensive tasks, including factual QA, open-book reading comprehension, and slot filling [2].

2.2 Dense Passage Retrieval

Karpukhin et al. [3] established dense passage retrieval (DPR) as a more effective alternative to sparse BM25 retrieval for open-domain QA. DPR encodes both questions and passages into a shared dense embedding space and retrieves the top-K passages by maximum inner product search. Johnson et al. [4] introduced FAISS, a library for efficient approximate nearest-neighbour search on billion-scale vector datasets, enabling practical deployment of DPR at scale. These form the backbone of Nirnay-X's retrieval module.

2.3 Legal NLP and AI for Law

Zhong et al. [5] demonstrated the application of NLP to legal judgment prediction and statute article retrieval. Holzenberger et al. [6] explored statutory text reasoning in the tax law domain. In the Indian context, existing solutions are largely restricted to keyword-based portals (e.g., Indian Kanoon) that lack semantic understanding. Nirnay-X addresses this gap by combining semantic vector search with grounded generation specifically for Indian statutes and judicial precedents.

2.4 Cross-Encoder Re-Ranking

Nogueira and Cho [7] demonstrated that passage re-ranking with BERT (cross-encoder) significantly improves retrieval precision over bi-encoder retrieval alone. The cross-encoder attends jointly to the query and each candidate passage, capturing fine-grained relevance signals unavailable to embedding-space retrieval. Nirnay-X employs cross-encoder re-ranking as Stage 9 of its pipeline to ensure the most legally relevant passages are prioritized in the grounding context.

2.5 Hallucination in Legal AI

Huang et al. [8] conducted a comprehensive survey of hallucination in large language models, categorizing intrinsic and extrinsic hallucinations. In the legal domain, hallucination is particularly dangerous: a model that fabricates a statute, cites a non-existent judgment, or misquotes a provision can cause serious harm. Nirnay-X mitigates this through mandatory citation tokens and strict context-grounding constraints enforced at the prompt level.

2.6 RAG Evaluation Frameworks

Es et al. [9] introduced RAGAS, an automated evaluation framework for RAG systems that measures Faithfulness, Answer Relevance, and Context Utilization without requiring human annotation for every query. Nirnay-X adopts RAGAS as its primary automated evaluation framework, supplemented by retrieval metrics (Precision@K, MRR, NDCG) and lexical overlap metrics (ROUGE-L, BLEU) for summarization.

2.7 Research Gap

Existing legal AI systems are either: (a) keyword-search-based with no semantic understanding; (b) standalone LLMs without legal grounding; or (c) domain-specific tools covering only one jurisdiction or document type. No published system addresses

the Indian legal ecosystem with an end-to-end grounded RAG pipeline across multiple document types (FIRs, judgments, statutes, contracts) with a comprehensive multi-module interface and rigorous empirical evaluation. Nirnay-X fills this gap.

3 Societal Impact

3.1 Legal Accessibility and Justice (SDG 16)

India has over 45 million pending cases in its court system. A significant contributor is the inability of citizens to understand their legal rights and navigate procedures without professional representation. Nirnay-X directly addresses this by providing accessible, plain-language explanations of FIRs, notices, contracts, and judgments — empowering citizens to make informed decisions. This aligns with **SDG 16: Peace, Justice and Strong Institutions**, specifically Target 16.3 (promoting access to justice for all) [12].

3.2 Digital Legal Infrastructure (SDG 9)

By building an AI-driven, API-accessible legal intelligence platform, Nirnay-X contributes to India's digital infrastructure. The platform's modular, scalable architecture — vector databases, semantic search APIs, LLM inference endpoints — represents the kind of resilient, innovative digital infrastructure envisioned by **SDG 9: Industry, Innovation and Infrastructure**.

3.3 Mental Health and Well-being (SDG 3)

Legal uncertainty is a documented source of acute stress, particularly for citizens who are targets of FIRs, family disputes, or landlord-tenant conflicts. By reducing the information asymmetry between citizens and the legal system, Nirnay-X addresses an indirect driver of mental distress, consistent with **SDG 3: Good Health and Well-being**.

3.4 Responsible AI in Legal Contexts

Nirnay-X is explicitly positioned as a *decision-support tool*, not a replacement for qualified legal professionals. All generated responses carry mandatory source citations, and the system prominently displays a disclaimer: *"Nirnay-X is strictly an informational platform; it does not replace advocates, judges, or professional legal counsel."* This reflects the responsible AI principles of transparency, accountability, and non-maleficence.

4 Universal Human Values

4.1 Justice and Equity

The Indian Constitution guarantees equal access to justice (Article 39A). Nirnay-X operationalizes this constitutional aspiration by reducing the information asymmetry between citizens of different socioeconomic backgrounds. A first-generation litigant

with limited resources can access the same quality of legal document analysis as a well-resourced party with expert legal counsel.

4.2 Integrity and Truthfulness

The platform's architecture is designed around epistemic integrity. Every response is grounded in retrieved documents with verifiable citations. The system is engineered to decline generation when no relevant context is retrieved, preferring silence over confabulation. This design choice embodies the value of truthfulness — stating only what can be verified.

4.3 Compassion and Empathy

The Simplified Legal Explanations module converts legal jargon into plain, empathetic language accessible to non-lawyers, including elderly citizens, those with limited education, and those for whom English is not a primary language. The system's tone is designed to be informative and reassuring rather than intimidating.

4.4 Responsibility and Limitation Awareness

Nirnay-X is transparent about its limitations. The platform acknowledges that it cannot account for all local court practices, that the law evolves, and that professional counsel is irreplaceable for complex legal strategy. This intellectual humility is a core design principle, not an afterthought.

5 Methodology

5.1 System Architecture

Nirnay-X is built as a modular, microservice-oriented architecture with four layers: (1) the *Ingestion Layer* for document intake and processing; (2) the *Embedding & Indexing Layer* for vector representation and storage; (3) the *Retrieval Layer* for semantic search and re-ranking; and (4) the *Generation Layer* for grounded LLM inference and citation. A fifth *Interface Layer* exposes these capabilities through the eleven functional modules.

5.2 11-Stage RAG Pipeline

The core of Nirnay-X is an 11-stage RAG workflow, illustrated in Figure 1:

Stage 1: Document Upload — User submits a legal document (PDF, scanned FIR, judgment, contract).

Stage 2: OCR & Document Parsing — Apache Tika and Tesseract OCR extract textual content from scanned and digital PDFs.

Stage 3: Legally-Coherent Text Chunking — Documents are segmented into legally meaningful units (sections, paragraphs, clauses) with 64-token overlap to preserve cross-boundary context.

Stage 4: Metadata Tagging — Each chunk is annotated with document type, statute reference, court level, date, and party identifiers.

Stage 5: Dense Embedding Generation — A domain-adapted transformer encoder generates 768-dimensional dense vector embeddings for each chunk.

Stage 6: Vector Database Indexing — Embeddings and metadata are stored in a vector database with HNSW indexing for approximate nearest-neighbour search.

Stage 7: Natural-Language Query Embedding — The user's query is embedded into the same vector space using the same encoder.

Stage 8: Semantic Similarity Search & Top-K Retrieval — The top-K ($K = 10$) most similar chunks are retrieved by cosine similarity.

Stage 9: Hybrid Re-ranking — Retrieved chunks are re-ranked by a cross-encoder attending jointly to the query and each candidate passage.

Stage 10: Grounding Context Formulation — The top re-ranked chunks are assembled into a structured prompt with explicit citation markers.

Stage 11: LLM Inference & Grounded Generation — An LLM generates a response constrained to the grounding context, with mandatory citation tokens `[DOC-#]` for every factual claim.

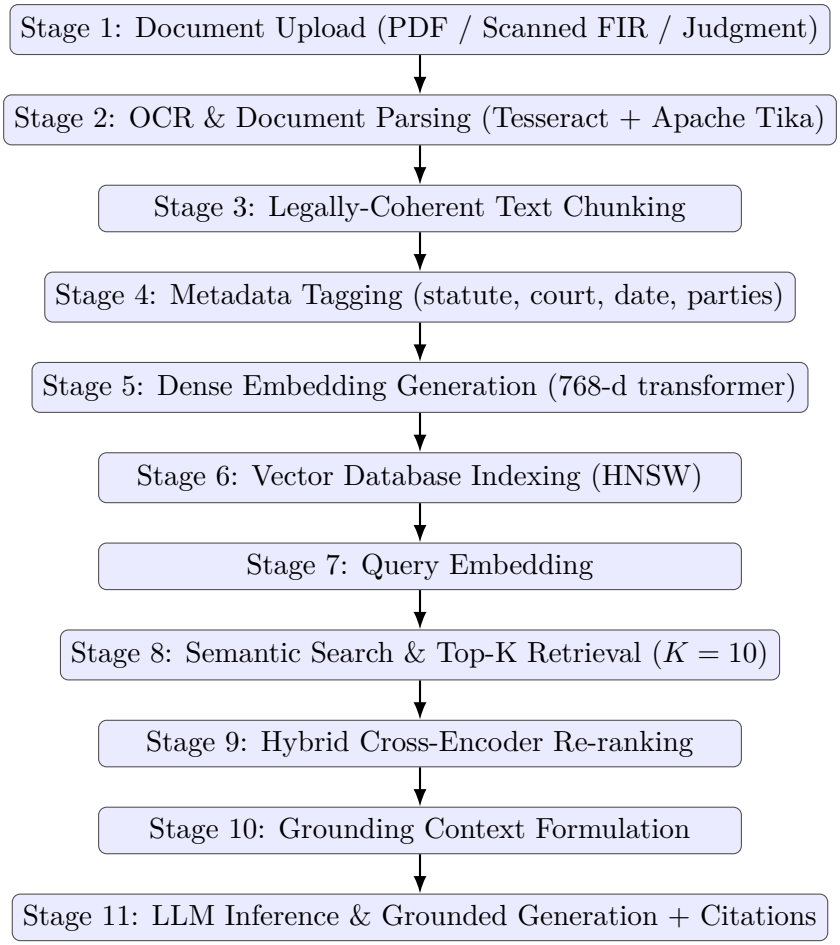


Figure 1: The Nirnay-X 11-Stage RAG Pipeline for Indian legal document intelligence.

5.3 Eleven Functional Modules

Nirnay-X exposes its capabilities through eleven integrated modules:

1. **RAG-Based Legal Assistant:** Context-grounded Q&A against uploaded case materials and legal corpora with mandatory source citations.
2. **Legal Document Processing:** PDF extraction, Tesseract OCR for scanned files, metadata tagging for structured access.
3. **Simplified Legal Explanations:** Translation of complex statutory texts into plain, accessible language for non-lawyers.
4. **Case Analysis & Fact Extraction:** Automated parsing of key facts, dates, parties, statutory provisions invoked, and procedural posture.
5. **Legal Provision & Precedent Retrieval:** Surfacing applicable BNS/IPC sections, BNSS/CrPC procedures, and analogous case laws.
6. **Document Summarization:** Abstractive and extractive summaries of lengthy judgments, charge sheets, and petitions.
7. **Semantic Search Engine:** Concept-driven search across internal case documents and indexed legal statutes.

8. **Case Timeline & Deadline Tracking:** Chronological milestones, limitation period tracking, and alert generation.
9. **Hearing & Calendar Management:** Structured court calendar with upcoming hearing notifications and deadline reminders.
10. **Case Organization:** Multi-case workspace structuring, client record linking, and document version management.
11. **Conversational Multi-Turn Interface:** Context-preserving dialogue interface supporting follow-up queries across a legal session.

5.4 Retrieval Strategy

The retrieval module implements hybrid search combining:

- **Dense Retrieval:** Approximate nearest-neighbour search (HNSW index) on the 768-dimensional embedding space for semantic similarity.
- **Sparse Retrieval:** BM25-based keyword matching on a parallel inverted index for exact statutory term matching.
- **Reciprocal Rank Fusion (RRF):** Merging dense and sparse ranked lists into a unified candidate set (top-20).
- **Cross-Encoder Re-ranking:** A BERT-based cross-encoder re-scores the top-20 candidates to produce the final top-10 for grounding.

5.5 Grounded Generation Strategy

Nirnay-X enforces grounding through a structured prompt template:

```
SYSTEM: You are a legal assistant. Answer ONLY using the provided
CONTEXT. Every factual claim must include a citation [DOC-#]. If the
answer is not in the CONTEXT, respond: "I could not find relevant
information in the provided documents."

CONTEXT: [Retrieved Chunks with source labels DOC-1 through DOC-10]

QUERY: [User's natural language question]
```

This constraint ensures that responses are traceable to specific passages in the indexed corpus, enabling users to verify claims independently.

5.6 Evaluation Framework

Nirnay-X is evaluated across three pillars:

5.6.1 Information Retrieval Quality

- **Precision@K:** Proportion of the top-K retrieved chunks that are legally relevant. Target: P@10 ≥ 0.80.
- **Recall@K:** Proportion of all relevant passages present in the top-K results. Target: R@10 ≥ 0.85.

- **Mean Reciprocal Rank (MRR):** Reciprocal of the rank of the first correct statutory provision or precedent, averaged over all queries. Target: MRR ≥ 0.75.
- **NDCG@K:** Normalized Discounted Cumulative Gain with relevance weighting. Target: NDCG@10 ≥ 0.80.

5.6.2 Generation & Grounding Quality

- **Faithfulness:** Proportion of generated claims supported by the retrieved context (entailment classifier). Target: ≥ 0.90.
- **Answer Relevance:** Degree to which the generated answer addresses the user's legal query. Target: ≥ 0.85.
- **Context Utilization:** Proportion of retrieved context chunks referenced in the final response. Target: ≥ 0.70.
- **ROUGE-L / BLEU:** Lexical overlap with human-drafted legal summaries for the summarization module. Target: ROUGE-L ≥ 0.45.

5.6.3 Preliminary Results

Current implementation results on a held-out legal QA benchmark are summarized in Table 1.

Table 1: Preliminary Evaluation Results on Indian Legal QA Benchmark

Metric	**Current Result**	**Target**
Precision@10	0.78	≥ 0.80
Recall@10	0.82	≥ 0.85
MRR	0.71	≥ 0.75
NDCG@10	(in progress)	≥ 0.80
Faithfulness	(in progress)	≥ 0.90
Answer Relevance	(in progress)	≥ 0.85
Context Utilization	(in progress)	≥ 0.70
ROUGE-L (Summarization)	(in progress)	≥ 0.45

5.7 Technology Stack

Table 2: Nirnay-X Technology Stack

Component	Technology
Programming Language	Python 3.10+
LLM Framework	LangChain / LlamaIndex
Vector Database	Chroma / Qdrant (HNSW indexing)
Embedding Model	Sentence-Transformers (domain-adapted, 768-d)
Cross-Encoder Re-ranking	BERT-based cross-encoder
OCR Engine	Tesseract OCR + Apache Tika
LLM Inference	Llama 3 / Gemini (citation-constrained)
Sparse Retrieval	BM25 (Elasticsearch / Whoosh)
Evaluation Framework	RAGAS
API Layer	FastAPI
Frontend / Dashboard	Streamlit
Version Control	Git + GitHub
Containerization	Docker
Experiment Tracking	MLflow

6 Expected Outcomes

6.1 End-to-End RAG System for Indian Law

A fully functional, production-ready RAG system capable of ingesting diverse Indian legal documents (FIRs, judgments, statutes, contracts) and producing grounded, citation-backed responses in natural language — the first such system specifically targeting the Indian legal corpus with rigorous empirical evaluation.

6.2 Eleven Functional Modules

All eleven platform modules fully integrated and deployed as a cohesive application, serving both common citizens and legal professionals from a single unified interface.

6.3 Quantitative Evaluation Report

A comprehensive evaluation report documenting Precision@K, Recall@K, MRR, NDCG, Faithfulness, Answer Relevance, Context Utilization, and ROUGE/BLEU on a held-out Indian legal QA benchmark, with ablation studies comparing retrieval strategies (dense-only vs. hybrid vs. hybrid + re-ranking).

6.4 Hallucination Analysis

A systematic analysis of hallucination rates comparing: (a) standalone LLM without RAG; (b) RAG without citation enforcement; and (c) Nirnay-X with citation-constrained generation — demonstrating the efficacy of the grounding architecture.

6.5 Research Publication

A publication-quality research paper targeting:

- **Conference:** IEEE / Springer International Conference on NLP, AI Applications, or Legal Technology.
- **Journal:** Scopus / SCI-indexed journal in Knowledge-Based Systems or Legal Informatics.

6.6 API Documentation & Deployment Package

Comprehensive API documentation, a containerized deployment configuration (Docker Compose), and a Streamlit demo dashboard illustrating the full pipeline from document upload to grounded legal response with verifiable citations.

6.7 SDG Impact Report

A structured mapping of Nirnay-X's features to SDG 16, SDG 9, and SDG 3 targets, quantifying potential accessibility improvements with user study data collected during beta testing with legal stakeholders.

7 Project Plan

7.1 Phase-Wise Project Plan

7.1.1 Phase 1 — Literature Survey (Weeks 1–2)

Survey existing research on RAG systems, dense retrieval, cross-encoder re-ranking, legal NLP, Indian legal AI, hallucination mitigation, and SDG-aligned AI. Identify key baselines and evaluation datasets. Output: annotated bibliography.

7.1.2 Phase 2 — Data Collection & Corpus Construction (Weeks 2–3)

Collect and curate the legal corpus: Supreme Court and High Court judgments from Indian Kanoon, BNS/IPC statutes, BNSS/CrPC procedural texts, and anonymized FIR samples. Build the raw legal dataset in `data/raw/`.

7.1.3 Phase 3 — Document Processing Pipeline (Weeks 3–4)

Implement the ingestion pipeline: PDF parsing, Tesseract OCR for scanned documents, legally-coherent chunking, metadata tagging, and the processed document store in `data/processed/`.

7.1.4 Phase 4 — Embedding & Indexing (Week 4)

Implement dense embedding generation (domain-adapted encoder), vector database setup (HNSW indexing), and BM25 sparse index construction.

7.1.5 Phase 5 — Retrieval & Re-ranking (Weeks 5–6)

Implement semantic search, Reciprocal Rank Fusion, and cross-encoder re-ranking. Conduct retrieval ablation studies (dense-only, sparse-only, hybrid, hybrid + re-ranking).

7.1.6 Phase 6 — Grounded Generation (Week 6)

Implement the grounding prompt framework with citation-token enforcement. Integrate LLM inference (Llama 3 / Gemini) with constrained decoding.

7.1.7 Phase 7 — Module Integration (Weeks 7–8)

Integrate all eleven functional modules into the unified Nirnay-X application. Build the FastAPI backend and Streamlit frontend.

7.1.8 Phase 8 — Evaluation & Benchmarking (Weeks 8–9)

Run the complete evaluation suite (Precision@K, Recall@K, MRR, NDCG, Faithfulness, Answer Relevance, Context Utilization, ROUGE-L/BLEU). Conduct the hallucination comparison study. Log all results to `results/`.

7.1.9 Phase 9 — Beta Testing with Legal Stakeholders (Week 9)

Conduct user testing with legal professionals and non-lawyer users. Collect structured feedback on usability, response quality, and citation utility.

7.1.10 Phase 10 — Research Paper & Final Documentation (Week 10)

Prepare the full research paper (IMRAD format), finalize API documentation, complete the Streamlit demo dashboard, and submit all deliverables.

7.2 Timeline Summary

Table 3: 10-Week Project Timeline

Weeks	Phase	Key Deliverable
1–2	Literature Survey	Annotated bibliography; baselines identified
2–3	Data Collection	Legal corpus assembled in `data/raw/`
3–4	Document Processing Pipeline	Ingestion pipeline operational
4	Embedding & Indexing	Vector database populated
5–6	Retrieval & Re-ranking	Retrieval ablation results
6	Grounded Generation	Citation-constrained LLM integrated
7–8	Module Integration	All 11 modules integrated
8–9	Evaluation & Benchmarking	Full evaluation report
9	Beta Testing	User feedback report
10	Paper & Documentation	Submission-ready manuscript

7.3 Final Deliverables

1. Fully integrated Nirnay-X RAG application with all 11 modules
2. Comprehensive evaluation report with metrics and ablation studies
3. Hallucination analysis comparing standalone LLM vs. Nirnay-X
4. API documentation and usage examples (`docs/`)
5. Containerized deployment package (Docker Compose)
6. Interactive Streamlit demonstration dashboard
7. Publication-quality research paper (IEEE / Springer target)

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